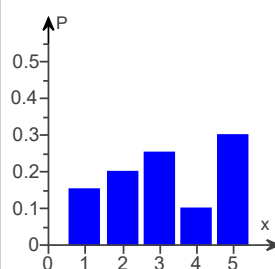


Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 8**Date:** 1/15/24**Course:** Calculus Beta

Find the expected value for the random variable whose probability function graph is displayed here.



The expected value of a probability distribution is the sum of the products of each of the x-values of the distribution and its probability.

$$E(X) = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots + x_n p_n$$

Start by determining the probability of each value of x. The probability that $x = 1$ is the height of the bar labeled 1.

The probability that $x = 1$ is 0.15.

The probability that $x = 2$ is 0.20.

The probability that $x = 3$ is 0.25.

The probability that $x = 4$ is 0.10.

The probability that $x = 5$ is 0.30.

Substitute each value of x and its probability into the formula for expected value.

$$\begin{aligned} E(X) &= x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5 \\ &= 1 \cdot 0.15 + 2 \cdot 0.20 + 3 \cdot 0.25 + 4 \cdot 0.10 + 5 \cdot 0.30 \end{aligned}$$

Simplify the expression.

$$\begin{aligned} E(X) &= 0.15 + 0.40 + 0.75 + 0.40 + 1.50 \\ &= 3.2 \end{aligned}$$

The expected value of the probability distribution is 3.2.